AI Assistants for Scientific Coding

AMSA 2026 workshop

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This session's roadmap

1. **AI tools for data analysis** - Background
2. **Setup** — Positron / VS Code + Copilot
3. **Section 1** — Line-by-line completion
4. **Section 2** — Agents
5. **Afternoon** — your choice (see the end of this deck)

> Tip
>
> Each activity is short — the book has full walkthroughs and a Challenge for every one.

Intro to LLMs for coding

A story



AI is replacing coding by hand

Almost everyone is using it now

> 22% of marine conservation professionals are using AI to help with data tasks. 34% want to use it

Source: octogroup.org

Serious analysts are using it even more

But is it any good?

Who's in the room?

Quick introductions

- Name
- Institution
- What your research is on

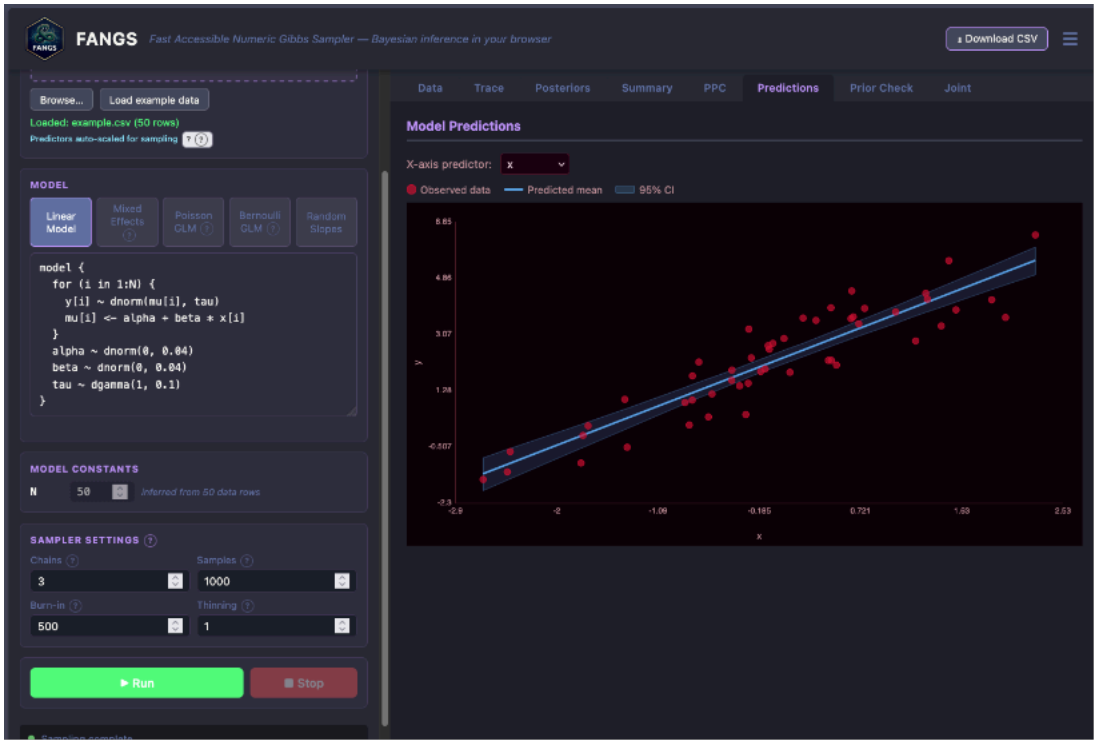
How do LLMs work?

Predict one token at a time

https://www.youtube.com/watch?v=wjZofJX0v4M

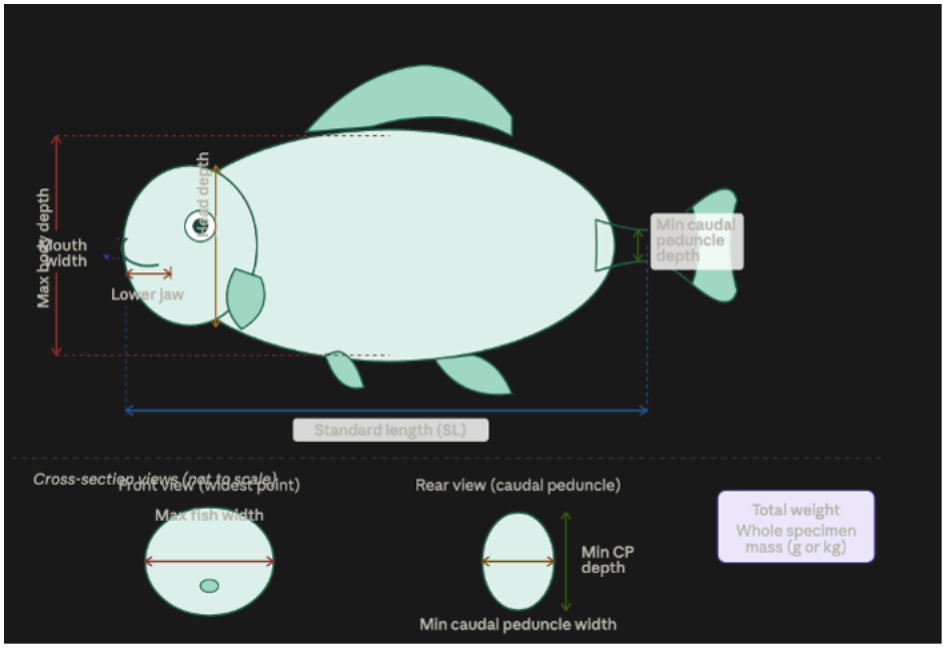
(skip to 1:25)

Simultaneously amazing...



FANGS - Chris' own Bayesian modelling software that runs in your browser

And underwhelming...



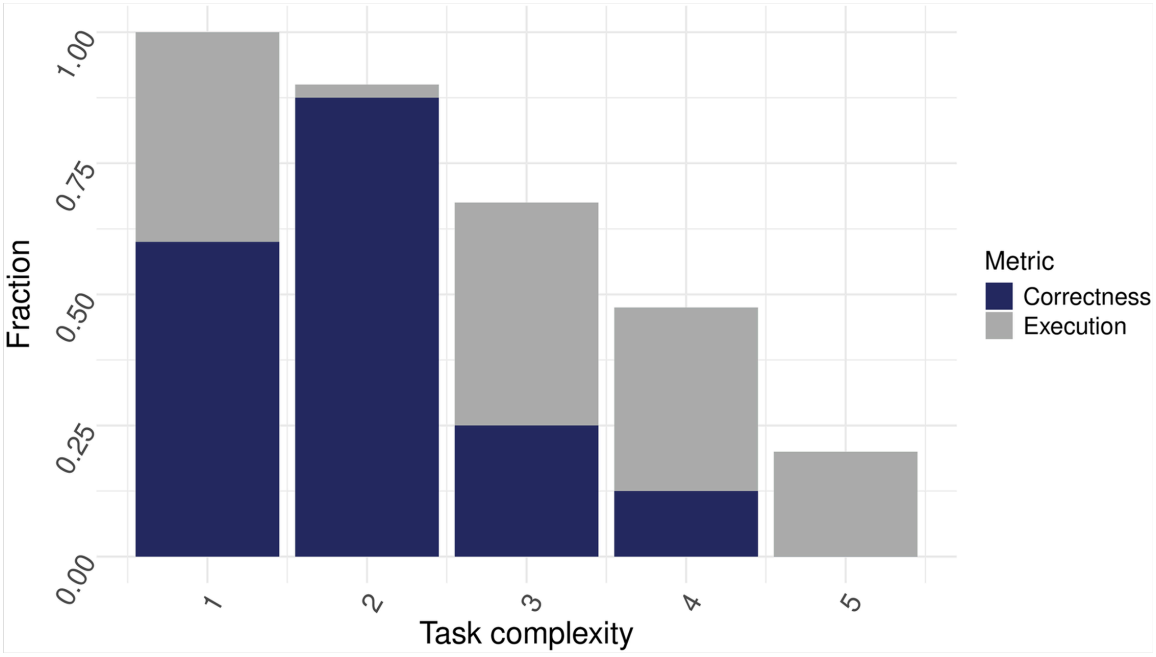
Claude 3.7 morphological diagram of a fish

Risks to scientific integrity?

Evaluating LLMs for coding

- Repeat the same prompt 10 times
- Score answers against a solution or rubric
- This is what software engineers do
- We can do it for data analysis too

Code that works, but is often scientifically flawed



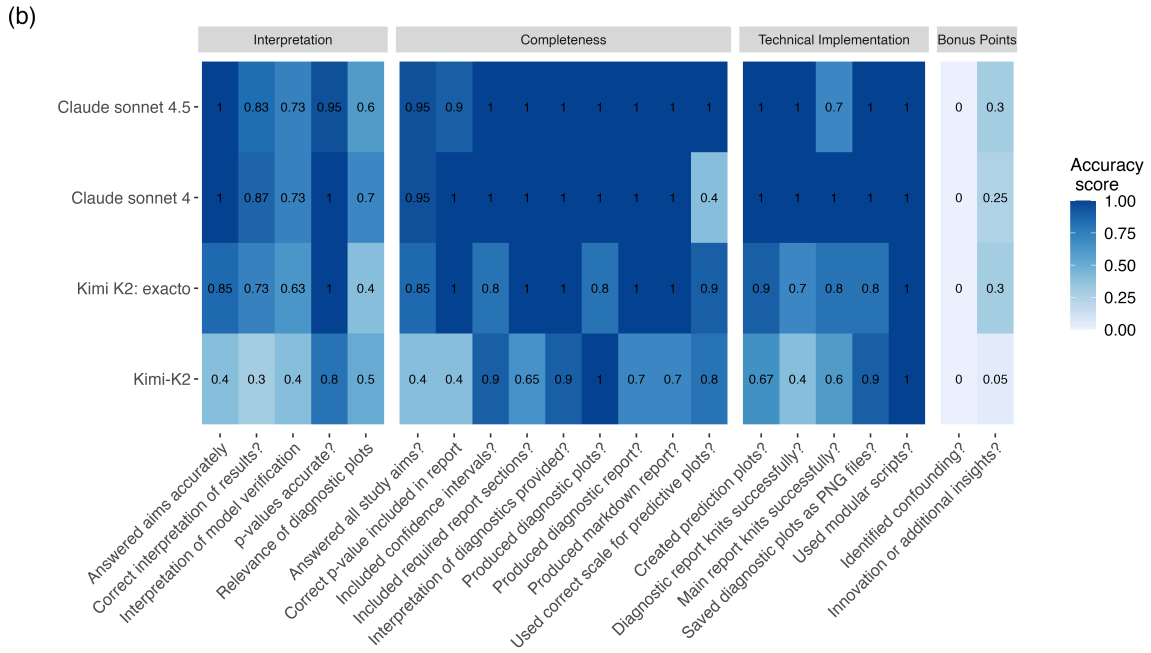
Jansen et al. 2025 “Leveraging large language models for data analysis automation””

Can AI agents complete an ecological analysis report?

- Paper: “Automating Ecological and Fisheries Modelling With Agentic AI”
- Ran AI agents with different LLMs
- Repeated each prompt 10 times
- Goal was to create a full data analysis with report
- Marked to a rubric

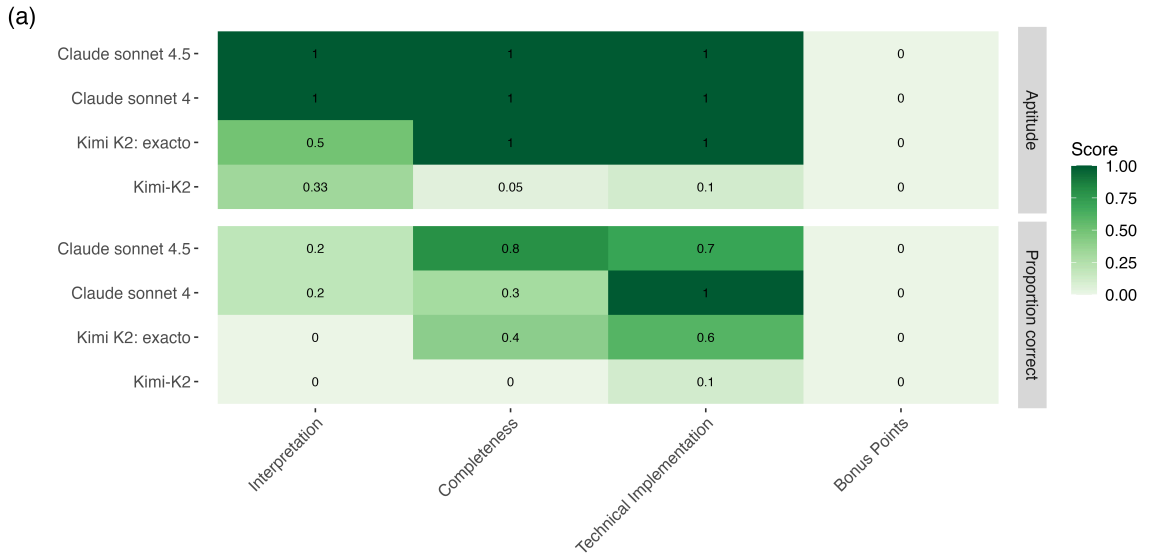


Results



Brown et al. 2026 “Automating Ecological and Fisheries Modelling With Agentic AI”

Results



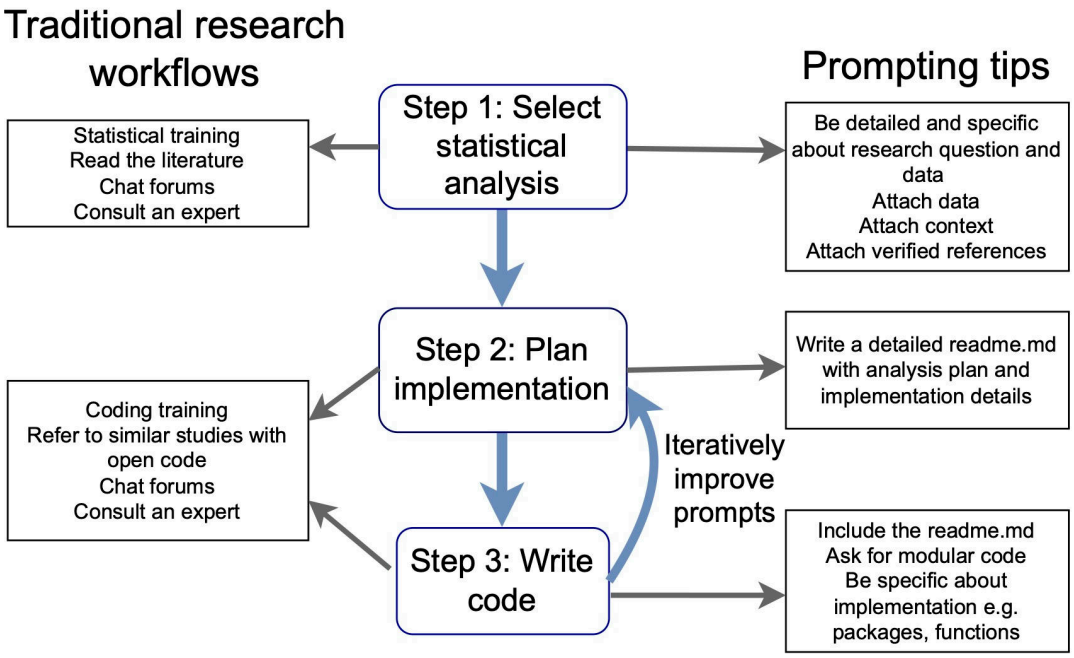
Brown et al. 2026 “Automating Ecological and Fisheries Modelling With Agentic AI”

How best to use AI coding tools?

Brown and Spillias 2026 “Prompting large language models for quality ecological statistics”



Break the problem into parts

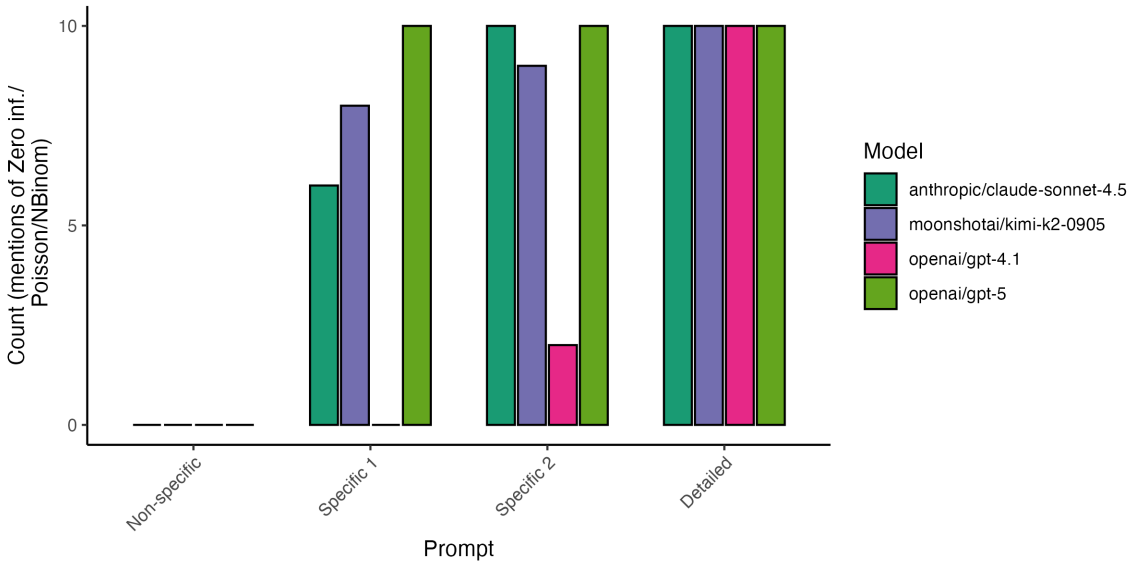


Brown and Spillias 2026 “Prompting large language models for quality ecological statistics”

1. Select analysis

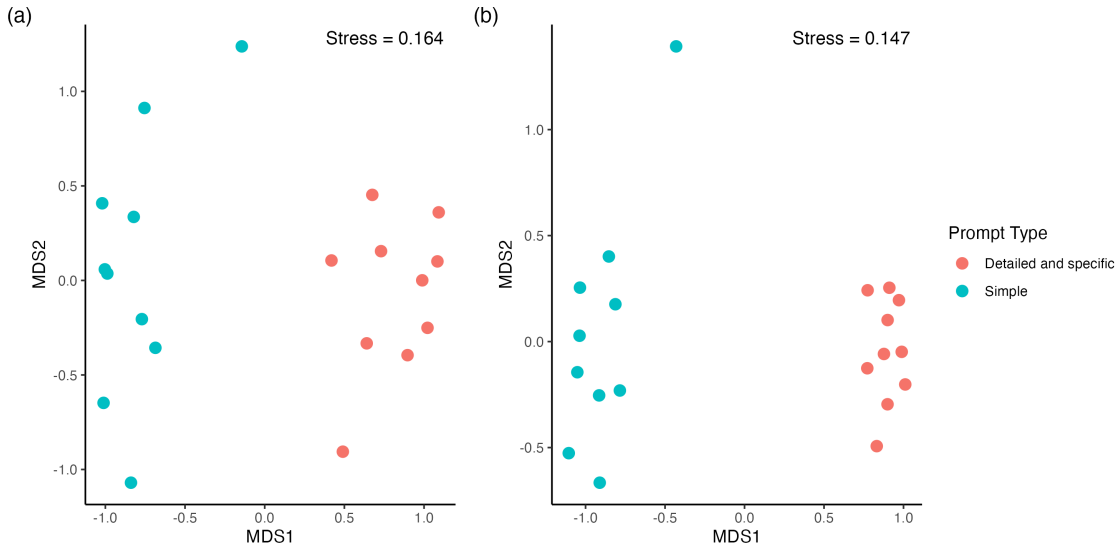
- Be specific
- Give detailed context
- Attach data
- Attach verified context (e.g refs)

1. Select analysis



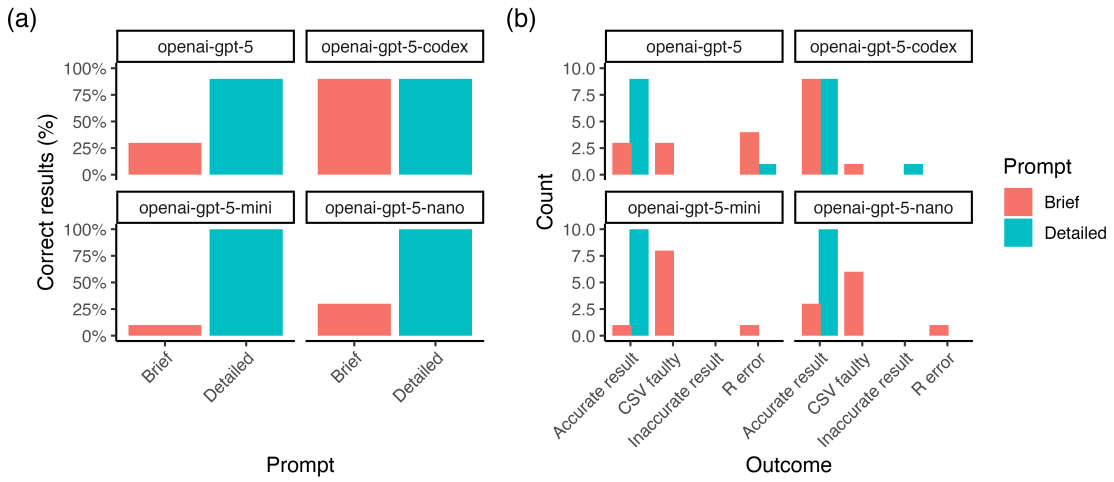
2. Plan implemenetation

More detailed plans give more consistent code



3. Write code

Detailed plans = more accurate results with cheaper models.



Today's workshop

We'll practice these skills

Also time for being flexible with learning goals

Setup

Getting your tools ready

Set up your IDE

Positron or VS Code, plus GitHub Copilot

1. Install Positron (easiest for R users) — or VS Code and its clones (Cursor, Windsurf)
2. Install the GitHub Copilot extension and sign in with your GitHub account
3. Using `ellmer` or Gander too? Get an API key — OpenRouter is the recommended default

Full step-by-step instructions: the book's Setup chapter.

Activity — Find your way around

Goal: spend a few minutes comparing your new IDE to RStudio

- Terminal / console — where R runs
- Script editor — where you write and edit code
- Environment / variables pane — the objects in your session
- Plots or viewer pane — where figures appear
- Extensions panel — where Copilot and other tools live

> ! Challenge
>
> Open a script, run a line of R, and find where your plot appears.

Get the data and prompts



Section 1

Line-by-line completion with GitHub Copilot

What is line-by-line completion?

- You type, and the assistant guesses what comes next
- Press `Tab` to accept — or keep typing to ignore it
- Best for small, targeted edits — mistakes are obvious

> **Tip**
>
> This is the “judicious” way to use AI: targeted, low-risk, and easy to verify as you go.

Section 1 overview →

Activity 1 — Inline completion

Goal: load the reef dataset and tab-complete a plot without typing the code yourself.

1. Load the benthic coral-cover & fish-count dataset
2. Start a new line, type `gg…` and watch the suggestion appear
3. Press `Tab` to accept, or keep typing to steer it

> **Challenge**
>
> Plot `CB_cover` against `dist_to_logging_km` without typing any variable names first — how many attempts before it uses the right columns?

~10 minutes. Full module →

Activity 2 — Completion via comment creation

Goal: write what you want in a comment, and let Copilot write the code underneath it.

1. Write a one-line comment above the cursor
2. Trigger the suggestion and press `Tab`
3. Try templating a whole sequence of plots from comments alone

> **Prompt**
>
> ```
> # Make a point plot of secchi against pres.topa
> ```

> **Challenge**
>
> Produce a grouped summary of `secchi` and `pres.topa` by `flow` — comments only, no typed R code.

~10 minutes. Full module →

Gander — an RStudio-friendly alternative

- Stays inside RStudio — no need to switch IDE
- Sees inside your R session, so it already knows your variable names and their types
- Select code, hit your shortcut key, and type a prompt

> **Live demo**
>
> "Make a ggplot of pres.topa against CB_cover, coloured by logged"
>
> "Add a stat_smooth to this plot"

Live demo, ~5 minutes. Details → — set-up instructions are in Setup.

Activity 3 — Next edit suggestions

Goal: write one working example, then let Copilot help generalise it into a function.

1. Write the code for a single case (e.g. one species)
2. Turn on Next Edit Suggestions (the octocat icon, bottom right)
3. Start wrapping it in a function — tab through each suggested edit

> **Why this matters**
>
> This is the same discipline an agent needs from you: a clear spec, worked out on one concrete case before you automate it. Next stop: Section 2.

~10 minutes. Full module →

Section 2

Agents — automate programming a whole workflow

How agents work

Agents run tools in a loop:

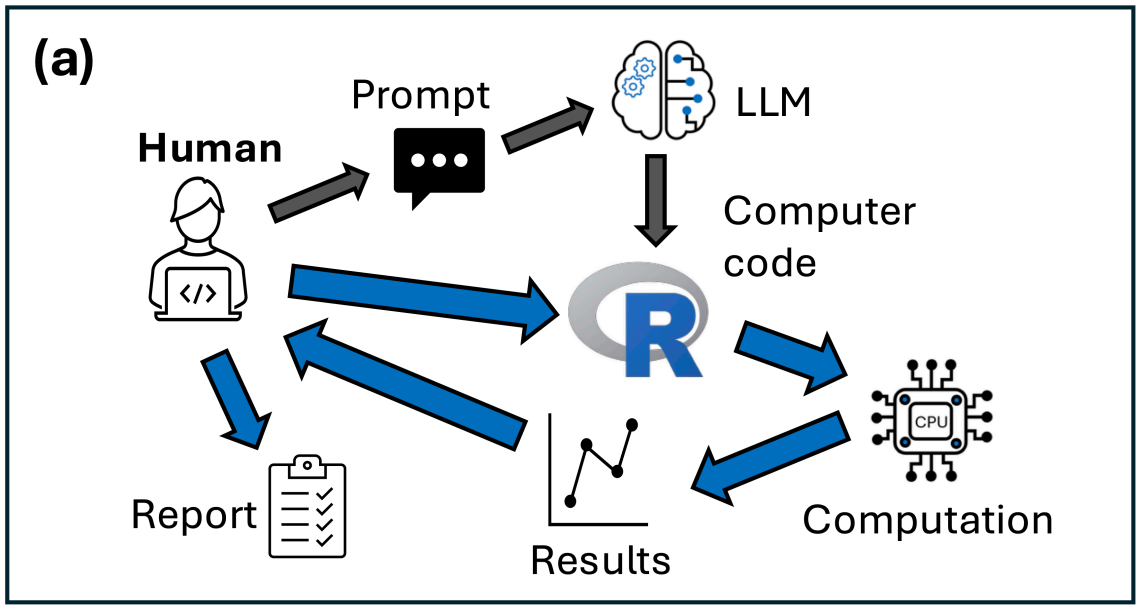
1. **Prompt**
2. **Model picks a tool**
3. **Tool runs**
4. **Result feeds back**

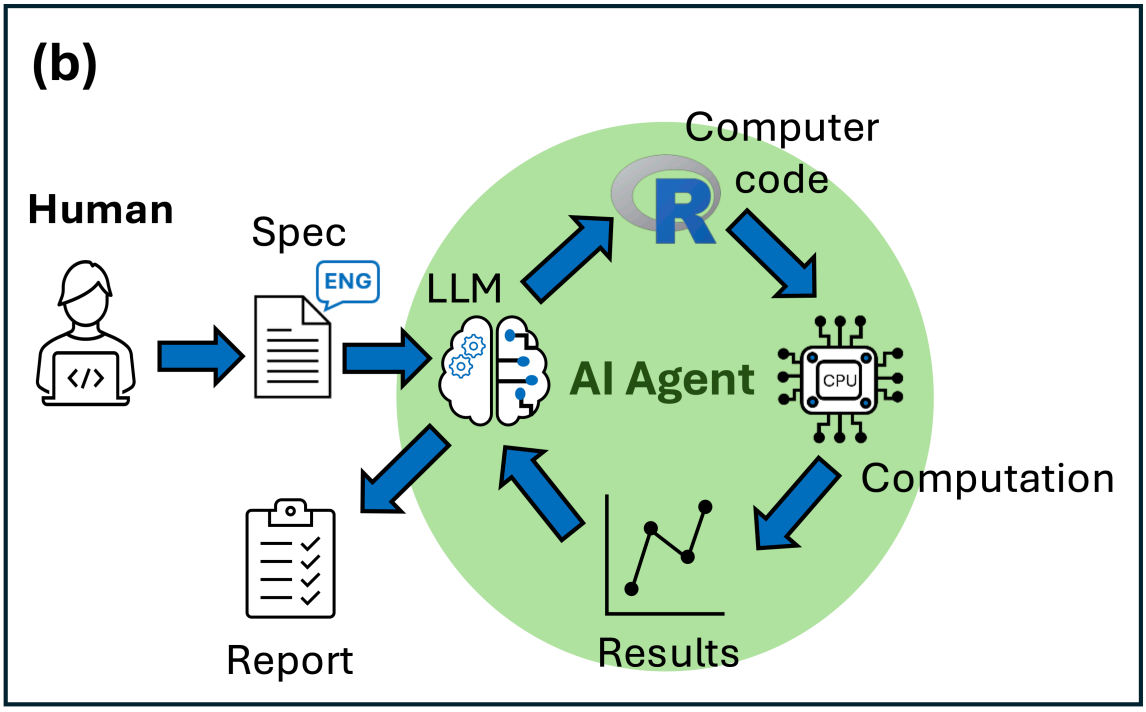
... and repeats until the model runs its 'task complete tool' or 'user question tool'

> **Note**
>
> By default, agents ask permission before each tool use. We'll unpack the security trade-offs properly in Section 4.

Section 2 overview →

How agents work





Brown et al. 2026 “Automating Ecological and Fisheries Modelling With Agentic AI”

Activity — Create an analysis with an agent

Goal: hand the agent a real regression task, end to end — scripts, run, and plots.

1. Switch your assistant to Agent mode

2. Open the data readme so the agent has variable context
3. Prompt it, then follow along — adjust or correct as it goes

> **Prompt**
>
> I want to do a regression of pres.topa on cb_cover. Create all the scripts I need, and run them to make plots of the predicted relationship.

> **Challenge**
>
> Run the same prompt twice, in two separate sessions — did the agent pick the same GLM family both times?

~15 minutes. Full module →

Prompting practice

Goal: keep an agent on track.

- Rich and relevant context is key
- Attach context: data summaries, metadata, domain knowledge
- Compare runs — same prompt, different agents or models

> **From the book**
>
> Try: "Arrr matey, I be wantin' a regression of pres.topa on cb_cover..."

~10 minutes. More on prompting agents → and part 2.

Specification practice

Goal: Agents do best with a detailed spec

Chris to show full spec for the benthic fish GLM.

Where that leaves us

3 completion activities

Inline, comments, and Gander

1 full agent run

A regression, end to end

2 prompting practices

Context, and content over style

Every activity has more detail, screenshots, and a Challenge in the book — go back any time: seascapemodels.org/AI-assistants-for-scientific-coding

What next?

Section 3 — AI in Workflows

Choose a statistical approach, plan it, then write the code

Section 4 — Security & Privacy

The risks behind "approve everything," and safer defaults like Claude for Web.

Section 5 — Customizing Agents

Turn what you learned into a reusable skill or sub-agent

Section 6 — Agentic Engineering

Testing & verification habits borrowed from software engineering

Section 7 — LLM Access via API

Go under the chat window — build a tiny agent yourself

Section 8 — Running Local Models

Models on your own machine, no API key required

Thank you

Book: seascapemodels.org/AI-assistants-for-scientific-coding

Blog: Vita ex Machina

Example data: github.com/cbrown5/example-ecological-data

